

Math 4441: Probability and Statistics

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Chapter 4
Conditional Distributions, Conditional Expectations and Independence
(Part 3 – Independence)
Lectures 23

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References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Independence

Two events A and B are independent if

$$P[AB] = P[A]P[B]$$

Two RVs X and Y are independent if, for any subsets A and B of real numbers, the events $\{X \in A\}$ and $\{Y \in B\}$ are independent,

$$P[X \in A, Y \in B] = P[X \in A]P[Y \in B]$$

Which can be easily extended, for any two real numbers x and y , to

$$P[X \leq x, Y \leq y] = P[X \leq x]P[Y \leq y]$$

If X and Y are two RVs defined on the same sample space, then X and Y are independent, iif, for all real numbers x and y ,

$$F_{XY}(x, y) = F_X(x)F_Y(y)$$

And is applicable for both discrete and continuous RVs

Independence of Discrete RVs

If X and Y are two discrete RVs with set of possible values S_X and S_Y , respectively, then they are independent, if

- $\forall x \in S_X$ and $\forall y \in S_Y$,
- the events $\{X = x\}$ and $\{Y = y\}$ are independent
- which, in turn, satisfies

$$P[X = x, Y = y] = P[X = x]P[Y = y] , \forall x \in S_X \text{ and } \forall y \in S_Y$$

$$P_{XY}(x, y) = P_X(x)P_Y(y)$$

In contrast, if X and Y are two discrete independent RVs with set of possible values S_X and S_Y , then

$$P[X = x|Y = y] = P[X = x]$$

i.e., knowing that $Y = y$, does not change the Probability of $\{X = x\}$

Example: Suppose there are 10 red and 30 blue balls in an urn. Suppose further that you first pick 13 balls from the urn with replacement, and then pick 20 balls with replacement. Find the probability that

- a) You pick 4 red balls among the first 13 balls and 7 red balls among the next 20 balls.
- b) You pick 11 red balls among the picked 33 balls

$X \triangleq$ # of red balls among the first 13 balls

$Y \triangleq$ # of red balls among the second 20 balls

$X \sim$

$Y \sim$

$P[X = 4, Y = 7] =$

Can be found from $P_{XY}(x, y)$

$$P_{XY}(4,7) = \binom{13}{4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^9 \binom{20}{7} \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^{13}$$

$$\begin{aligned} P_{XY}(x,y) &= \binom{n}{x} (p)^x (1-p)^{n-x} \binom{m}{y} (p)^y (1-p)^{m-y} \\ &= \binom{n}{x} \binom{m}{y} p^{x+y} (1-p)^{(n+m)-(x+y)} \end{aligned}$$

If X and Y are two independent RVs, then for real valued functions

$$g: R \rightarrow R \text{ and } h: R \rightarrow R$$

$W = g(X)$ and $Z = h(Y)$ are also independent RVs.

If X and Y are two independent RVs, then for real valued functions

$$g: R \rightarrow R \text{ and } h: R \rightarrow R$$

$$E[g(X)h(Y)] = E[g(X)]Eh(Y)]$$

A special case of which

$$E[XY] = E[X]E[Y]$$

However, the converse is not true. Above equation does not guarantee the independence of X and Y

Example: X is a RV with PMF $P_{X(x)} = \frac{1}{3}$, for $x = -1, 0, 1$, and $Y = X^2$, then prove that $E[XY] = E[X]E[Y]$ and X and Y are not independent, i.e.,

$$P_{XY}(x, y) = P_X(x)P_Y(y)$$

Independence of Continuous RVs

If X and Y are jointly continuous RVs with the JPDF $f_{XY}(x, y)$, then they are independent, iif

- $\forall x \in S_X$ and $\forall y \in S_Y$
- The JPDF is equal to the product of the marginal PDFs, $f_X(x)$ and $f_Y(y)$
- $f_{XY}(x, y) = f_X(x)f_Y(y)$

Example: Suppose there are two paths, A and B , between two computer nodes in a network. The PDF of the delays of packets, in hundreds of microseconds, for the two paths, is given by

$$f(x) = \begin{cases} \frac{x}{4}, & 1 < x < 3 \\ 0, & \text{otherwise.} \end{cases}$$

If the packet delay of one path is independent of the other, find the probability that, for two packets sent in two paths, the delay of one path is at least 50 microseconds more than the other path.

$X \triangleq$ delay of path A

$Y \triangleq$ delay of path B

The desired probability is

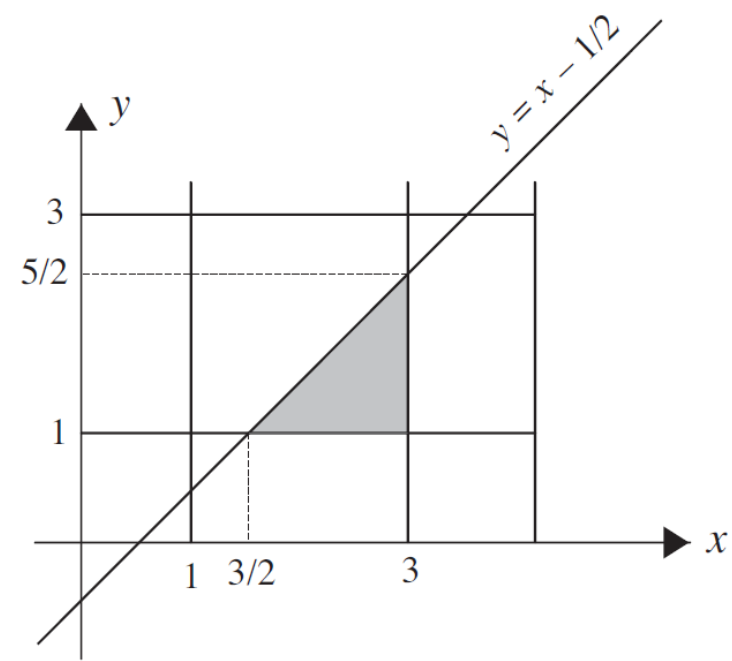
$$P\left[X > Y + \frac{1}{2}\right] + P\left[Y > X + \frac{1}{2}\right]$$

$$f_{XY}(x,y) = f_X(x)f_Y(y)$$

$$f_X(x) =$$

$$f_Y(y) =$$

$$f_{XY}(x,y) =$$



$$\begin{aligned}
2P\left(X > Y + \frac{1}{2}\right) &= 2P\left((X, Y) \in \left\{(x, y) : \frac{3}{2} < x < 3, 1 < y < x - \frac{1}{2}\right\}\right) \\
&= 2 \int_{3/2}^3 \left(\int_1^{x-1/2} \frac{xy}{16} dy \right) dx = \frac{1}{8} \int_{3/2}^3 \left[\frac{xy^2}{2} \right]_1^{x-1/2} dx \\
&= \frac{1}{16} \int_{3/2}^3 x \left[\left(x - \frac{1}{2}\right)^2 - 1 \right] dx \\
&= \frac{1}{16} \int_{3/2}^3 \left(x^3 - x^2 - \frac{3}{4}x \right) dx \\
&= \frac{1}{16} \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{3}{8}x^2 \right]_{3/2}^3 = \frac{549}{1024} \approx 0.54. \quad \blacklozenge
\end{aligned}$$

